

1. Administrative & Study Identification

Full Study Title: A Phase IIIB, Multinational, Multicenter, Randomized, Open-Label Study to Evaluate Patient Preference for Home Administration of Fixed-Dose Combination of Pertuzumab and Trastuzumab for Subcutaneous Administration in Participants With Early or Locally Advanced/Inflammatory HER2-Positive Breast Cancer **Short Title/Acronym:** ProHer **Posting ID:** 271270155124067012 **Protocol Number:** M043110 **NCT Number:** NCT05415215 **Sponsor/Company Name:** Hoffmann-La Roche (Roche) **Investigational Product:** Fixed-Dose Combination of Pertuzumab and Trastuzumab for Subcutaneous Use (PH FDC SC) **Phase of Development:** Phase 3 (Phase IIIB) **Trial Status:** COMPLETED **Preferred UMLS Name:** Patient Preference **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate patient preference for the fixed-dose combination of pertuzumab and trastuzumab for subcutaneous use (PH FDC SC) administration in the home setting compared with the hospital setting during the cross-over period of adjuvant treatment. **Secondary Objectives:** To assess the duration of treatment preparation, the perceived impact on clinical management/efficiency, and the perception of time/resource use by Healthcare Professionals (HCPs) via questionnaire, as well as to assess overall safety and pharmacokinetics.

2.2 Background & Rationale To address the logistical burden of intravenous (IV) cancer therapy for patients with HER2-positive breast cancer. Subcutaneous (SC) delivery of biotherapeutics is a well-established method that reduces administration time from hours to minutes. This study aims to demonstrate that at-home SC administration of dual-HER2 targeted therapy is safe, preferred by patients over hospital administration, and can improve the overall treatment experience and patient quality of life.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Cross-over design: Home setting vs. Hospital setting) **Blinding:** Open-label **Randomization:** 1:1 Stratified Randomization (Arm C: Hospital then Home vs. Arm D: Home then Hospital)

3.2 Planned Enrollment & Duration Target **NS**: 346 patients **Number of Sites**: 48 multinational locations **Disease Areas**: Early Breast Cancer, Locally Advanced Breast Cancer, Inflammatory Breast Cancer **Estimated Study Start**: 05-Jul-2022 **Estimated Primary Completion**: 20-Nov-2024 (Status: COMPLETED)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with Eastern Cooperative Oncology Group (ECOG) performance status 0-1. 2. Female and male participants with stage II-IIIC early or locally advanced/inflammatory HER2-positive breast cancer (primary tumor ≥ 2 cm in diameter, or node-positive disease). HER2+ status confirmed (3+ by IHC or positive by ISH). 3. Left ventricular ejection fraction (LVEF) $\geq 55\%$ by echocardiogram (ECHO) or MUGA, and intact skin at the planned site of SC injections. 4. Negative HIV, hepatitis B surface antigen (HBsAg), and hepatitis C virus (HCV) antibody tests at screening. 5. **For Treatment with Adjuvant PH FDC SC**: Completed the neoadjuvant phase, underwent surgery, and achieved pathologic complete response (pCR), defined as eradication of invasive disease in the breast and axilla.

4.2 Exclusion Criteria 1. Stage IV (metastatic) breast cancer. 2. Pregnancy, breastfeeding, or intending to become pregnant. 3. Renal function with creatinine clearance < 50 mL/min, inadequate bone marrow function, impaired liver function, serious cardiac illness, or history/risk factors for ventricular dysrhythmias. 4. Previous invasive non-breast cancer (unless disease-free for ≥ 5 years) or active unresolved infections requiring treatment. 5. Recent episode of thromboembolism with unoptimized anticoagulation dose and/or un-normalized INR.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: PH FDC SC (Trastuzumab and Pertuzumab) administered as a fixed, non-weight-based dose once every 3 weeks (1 cycle = 3 weeks). During the crossover period, participants receive 3 cycles in the hospital setting and 3 cycles in the home setting. **Route of Administration**: Subcutaneous (SC) injection. **Duration of Treatment**: Up to 18 total cycles of HER2-directed therapy (combining neoadjuvant, crossover, and continuation periods).

5.2 Concomitant & Prohibited Therapy Permitted: Investigator's choice of chemotherapy (during the neoadjuvant phase). **Prohibited**: Hormonal contraceptives (regardless of hormone receptor status); investigational therapy within 28 days prior to initiation. Radiotherapy must be fully completed prior to entering the cross-over period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, LVEF assessment, HIV/HBV/HCV Labs
Baseline	Day 0	Randomization into cross-over sequence (Arm C or D) post-surgery and post-radiotherapy
Interim	Cycles 1-6 (Cross-over)	Subcutaneous administration (Home/Hospital), Safety Labs, Administration of Patient Preference Questionnaire (PPQ)
End of Study	Up to Cycle 18	Final safety assessments, treatment continuation period (in patient's setting of choice), Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint 1. Patient Preference: Percentage of Participants Who Preferred the Administration of PH FDC SC in the Home Setting Compared With the Hospital Setting, based on Question 1 of the Patient Preference Questionnaire.

7.2 Secondary & Exploratory Endpoints

- 1. Duration of Treatment Preparation:** According to Healthcare Professionals' Responses to Question 1 of the Healthcare Professional Questionnaire (HCPQ) - Neoadjuvant Phase Drug Preparation Area.
- 2. Clinical Management & Efficiency:** Percentage of Healthcare Professionals by Their Responses on Perception of Impact of PH FDC SC on Clinical Management and Clinical Efficiency, Question 2 of the HCPQ - Neoadjuvant Phase Drug Preparation Area.
- 3. Time/Resource Use:** Percentage of Healthcare Professionals by Their Responses on Perception of Time/Resource Use of Each Study Regimen, Questions 3 and 4 of the HCPQ - Neoadjuvant Phase Drug Preparation Area.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via HCP assessment during hospital visits and home-care nursing visits.

Serious Adverse Events (SAEs): Safety monitoring including cardiac safety (LVEF monitoring) and administration-site reactions; standard 24-hour reporting timelines.

Data Monitoring Committee (DMC): Internal safety monitoring managed by Hoffmann-La Roche.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Adjuvant Cross-over Modified Intention-to-Treat Analysis Set (mITTcross), defined as all randomized patients who received at least one PH FDC SC treatment in the cross-over period and provided a valid response to PPQ Question 1.

Sample Size Justification: Target $N=346$ powered to evaluate superiority/preference of home administration over hospital

administration. **Handling of Missing Data:** Patients with missing responses to Question 1 of the PPQ, or those who incorrectly received radiotherapy after the start of the cross-over period, are excluded from the main mITTcross analysis.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Strict monitoring of protocol adherence, notably ensuring completion of post-surgery radiotherapy prior to cross-over initiation. **Audit Trail:** Secure eCRF locking and tracked, validated capture of patient-reported outcomes via the PPQ and HCPQ.

11. Study Identifiers & Governance

Primary ID: M043110 **Posting ID:** 271270155124067012 **Registry Number:** NCT05415215 **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche) **Study Title:** ProHer **Official Regulatory Title:** A Phase IIIB, Multinational, Multicenter, Randomized, Open-Label Study to Evaluate Patient Preference for Home Administration of Fixed-Dose Combination of Pertuzumab and Trastuzumab for Subcutaneous Administration in Participants With Early or Locally Advanced/Inflammatory HER2-Positive Breast Cancer

12. Clinical Framework (Breast Cancer Specific)

Primary Condition: Early or Locally Advanced/Inflammatory HER2-Positive Breast Cancer (Stage II-IIIC) **Diagnosis Threshold:** HER2+ status (3+ by IHC or positive by ISH amplification) **Medical Methodology:** Dual HER2-blockade via Subcutaneous Delivery (Trastuzumab/Pertuzumab) **Optimization Target:** Patient preference and adherence to home-based administration

13. Study Procedures & Methodology

Management Pathway: Subcutaneous cross-over clinical pathway (Hospital $\rightarrow$ Home) **Education Protocol (HCP):** Subcutaneous injection protocols and HCP questionnaire completion (HCPQ) **Education Protocol (Patient):** Home-care coordination and completion of Patient Preference Questionnaire (PPQ) **Quality Audit Mechanism:** Adherence to cross-over schedule and tracking of patient-reported outcomes

14. Observation & Follow-Up Schedule

Total Duration: Up to 18 cycles (approx. 54 weeks) plus safety follow-up **Onsite Visit Cadence:** Every 3 weeks (Q3W) during the hospital-based arm of the cross-over **Remote/Home Monitoring Frequency:** Every 3 weeks (Q3W) during the home-based arm of the cross-over **Checklist Review Interval:** At the end of the 6-cycle cross-over period (PPQ and HCPQ assessment)

15. Sign-off & Approval

Role	Name	Signature	Date						
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]						
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]						
Statistician	[Name]	_____	[DD-MMM-YYYY]						